

Abdullah Alshenaifi

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PROFESSIONAL SUMMARY

Applied Electronics and Control Systems Engineer and IEEE Professional Member, accredited by the Saudi Council of Engineers (SEC). Co-owner of Alamat Robotics, leading PCB design, hardware prototyping, and firmware for custom robotics and control systems. Experienced in production-line maintenance and fault diagnosis on automated inspection and palletizing equipment, and in teaching mechatronics and embedded systems at university level.

EXPERIENCE

Co-Owner & Technical Partner — Alamat Robotics, Riyadh *Apr 2019 – Present*

- Co-founded a robotics company (originating as a student team, now registered), delivering robotics, embedded systems, and PCB design for prototypes and MVPs.
- Lead electronics and PCB design, hardware prototyping, and firmware for custom robotics mainboards and control systems; mentor student teams in robotics and automation.

Lecturer — Mechatronics & Applied Electronics — Princess Nourah Bint Abdulrahman University (PNU), Riyadh *Jan 2025 – Jun 2026*

- Taught mechatronics and Arduino-based embedded systems to 100+ students across multiple sections, covering control systems, sensors and actuators, and automation labs.

Electronics Engineer — The National Company for Glass Industries (Zoujaj), Riyadh *Dec 2024 – Dec 2025*

- Maintained and troubleshooted electrical and electronic systems on production-line equipment, including TIAMA vision-inspection machines (MX4, MULTI4, MCAL4) and MSK palletizing and robotic systems.
- Diagnosed faults across sensors, actuators, drives, and control circuits to reduce downtime on active production lines.
- Built an internal maintenance-logging tool (Python, Streamlit) to replace manual paper logs with structured digital records.
- Developed a standardized reporting tool (React, Node.js) for the Electromechanical Department, consolidating shift data into a single reporting format.

Instructor — Robotics & Embedded Systems — Tuwaiq Academy, Riyadh *Jul 2024 – Aug 2024*

- Delivered an intensive Arduino, robotics, and IoT training curriculum; mentored student teams who won gold medals at IITEX.

Co-op Trainee — Electronics & PCB — King Abdulaziz City for Science and Technology (KACST), Riyadh *Jan 2021 – Apr 2021*

- Trained on PCB design, SMT/THT assembly workflows, and sensor integration in a research and development setting.

PROJECTS

IoT Smart Parking Management System — 2023

Embedded system for automated parking space detection and status reporting using sensor nodes and a wireless dashboard.

ESP32 | Embedded C | MQTT | IoT | Sensor Integration | Wireless Communication

Smart Safety Helmet with Fall Detection & Temperature Monitoring — April 2024

ESP32-based wearable with GPS tracking, temperature sensing, fall detection, and wireless Telegram alert notifications.

ESP32 | Embedded C | BLE | Wi-Fi | GPS | Hardware Design | Firmware

IoT Water Monitoring System — March 2023

Graduation project for apartment-level water metering with live dashboards and SMS alerts. Full-stack IoT build covering electronics, embedded programming, and cloud integration.

ESP32 | Embedded C | IoT | Sensors | Cloud Dashboard | Testing

Custom Robotics Mainboard PCB — August 2021

Custom robotics mainboard with complete schematic capture and PCB layout. github.com/AAShenaifi/Robots_MainboardPCB

PCB Design | Schematic Capture | Hardware Integration | EasyEDA | Documentation

SKILLS

Embedded Systems: ESP32, Arduino, STM32 (familiar), Embedded C/C++, Firmware Development
Connectivity & Protocols: MQTT, BLE, Wi-Fi, LoRa/LoRaWAN (familiar), UART / SPI / I2C
Hardware: PCB Design, Schematic Capture, SMT/THT Assembly & Rework, Sensor/Actuator Integration, Circuit Debugging
Industrial & Consulting: Automated Inspection Systems, Calibration/Alignment, Test & Verification/Validation, Engineering Studies & Documentation, Training Delivery
Software & Tools: Python (Streamlit), EasyEDA, React, Node.js, Technical Documentation

EDUCATION

Bachelor's Degree – Applied Electronics and Control Systems Engineering | Technical College of Riyadh | 2023
Diploma – Industrial Electronics and Control | Technical College of Riyadh | 2021

CERTIFICATIONS & TRAINING

- Training of Trainers (ToT) | Sahm Training Center | 2025
- Engineering Solutions Design and Manufacturing | Tuwaiq Academy | 2024
- Microelectromechanical Systems Technologies (MEMS) | Academy 32 | 2024
- Internet of Things (IoT) | Tuwaiq Academy | 2024
- 3D Printing | Tuwaiq Academy | 2024
- PLC Programming | SME | 2021
- Full Stack Engineer | Smart Methods | 2021

PROFESSIONAL MEMBERSHIPS

- Saudi Council of Engineers (SEC) | Electronics Engineer Accreditation | ID 1064201 | Sep 2024 – Present
- IEEE | Professional Member | ID 101800416 | Dec 2020 – Present

LANGUAGES

Arabic: Native | English: Advanced